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A review on the impact of big data analytics in transforming agricultural practices, food processing, and preservation strategies

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ABSTRACT

Information systems and big data analytics (BDA) are becoming increasingly vital for finding solutions to challenging problems in food and agriculture. Big data is used a lot in food distribution and retail, but not enough in production, making decisions, and making things last. BDA employs advanced techniques like machine learning and artificial intelligence to aggregate multiple data types, such as sensor data, satellite photos, and weather forecasts. It then looks for usable information that may be utilised to improve farming, food processing, and preservation. This systematic review brings together the most recent developments and highlights trends, issues, and areas where more research is needed on applying BDA in the agri-food sector. It highlights how BDA may improve quality assurance, precision agriculture, supply chain optimisation, and sustainable food systems. The main challenges are related to how data can be shared across systems, problems with infrastructure, privacy concerns and differences in how quickly stakeholders adopt these technologies. Therefore, the review suggested that future research should focus on interdisciplinary collaboration, addressing ethical considerations, and developing robust data governance frameworks to fully realise big data's potential in transforming food's future. The ethical and regulatory frameworks need to keep pace with advances in technology.

1. Introduction

Information systems have enabled businesses and society to solve complex challenges using data analysis for decision-making, predictions and strategy-making (Lioutas & Charatsari, 2020; Zhang et al., 2021). In a corporate environment, big data helps speed manufacturing, improve products and allow for quick decision-making (Li et al., 2022; Rathore et al., 2021). Nonetheless, in the food sector, big data is widely applied to distributing and retailing food but with limited involvement in producing, setting policies and dealing with sustainability issues. Big Data refers to the massive volumes of structured and unstructured data generated continuously from various sources. In contrast, Big Data Analytics (BDA) is the process of examining, processing, and analysing big data using advanced techniques such as machine learning, AI, and statistical methods to extract meaningful insights, patterns, and predictions that support informed decision-making (Jamarani et al., 2024). Ahmed and Shakoor (2025) reported that incorporating BDA in agriculture has created a significant opportunity to protect food systems, educate policy leaders, and resolve various social and environmental problems. When agricultural data is studied, the industry can learn about the links between making food and sustaining human and environmental welfare (Pawlak & Kołodziejczak, 2020).

Big data is essential in various agricultural sectors, including plant breeding, waste management, policy development, regulatory adherence, and supply chain optimisation (Rejeb et al., 2022; Sharma et al., 2021). Its implementation enhances traceability, food safety, and resource efficiency, concurrently diminishing food waste and the environmental impact of food production. The use of enormous amounts of data from sources including sensors, satellite imagery and weather projections is primarily driving this growth (Raji & Njoku, 2024). Aijaz

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et al. (2025) reported that farmers utilising these modern techniques to collect data could evaluate crop health, soil conditions, and weather variability to make informed decisions that improve production and reduce risk.

Furthermore, big data seeks to determine what we can figure out from the known. Data collection involves a wide range of sources, including sensors, remote sensing, GIS, GPS, RFID, yield mapping systems, and socio-economic data. These kinds of data that are often unstructured and similar to unstructured types are also known as "big data" (Mohamed et al., 2020; Naqvi et al., 2021). BDA, an excellent technique to convert big data structures into a usable form, has played a vital role in generating insights. Since big data can manage various formats, is designed for faster analysis, and is very comprehensive, it is increasingly expected to result in a revolution in Internet-based services and computing (Akhter & Sofi, 2022; Kumar et al., 2022). The ability to analyse big data has led to advancements in precision agriculture, allowing farmers to make more informed decisions based on real-time data. This has the potential to optimise resource allocation, increase crop yields, and ultimately improve overall efficiency in the agricultural sector.

Big data helped bring about precision agriculture that allows farmers to treat crops according to their needs, use inputs appropriately, and use modelling to predict what will happen to crops and pests (Bhat & Huang, 2021; Padhiary et al., 2024). Likewise, food processing companies use big data technologies to operate efficiently, ensure quality, and trace the process along the supply chain. Technology is used to check the storage and transport environment to maintain perishable products longer and help prevent spoilage (Duan et al., 2024).

Despite much research on BDA in agriculture and food systems, there is still no complete overview of its results. As a result, a systematic review will help integrate various studies, provide important direction for practical use, highlight areas for further research, and aid in the sustained transformation of agriculture and food systems by using big data (Gamage et al., 2024; Rejeb et al., 2022). Also, integrated frameworks that combine data collection, real-time analysis, and automation from production to consumption are not widely used. This kind of development prevents the construction of intelligent food systems working as one unit. There is a lack of research that truthfully compares the pros and cons of several BDA and AI tools and considers the challenges of using them in communities with less money and development. The most notable issues in agricultural sectors are data interoperability, poor infrastructure, and resistance to using technology, which are often identified but not thoroughly explored.

With this in mind, this systematic literature review is focused on exploring how big data is bringing positive changes to agriculture, food processing, and the way foods are stored and preserved. The report shows which trends, challenges, and research gaps exist, helping to prepare for further collaboration between scholars and practical use in projects. The objectives are as follows: (1) Look into using big data in modern farming. (2) Determine how big data helps improve food processing and maintain a high level of food quality. (3) See what obstacles and drawbacks prevent farmers from using big data technology throughout the supply chain.

As a result of this objective, the study looks to answer the following research questions. How are modern farmers leveraging BDA to achieve better productivity and sustainability? (2) How does BDA make a difference in the food processing industry's quality control and supply chain management? (3) What are some of the biggest challenges and problems that prevent proper adoption and use of BDA? (4) What parts of BDA still need agriculture, food processing, and preservation research, and what direction may it take?

Consequently, this systematic literature review will describe how BDA is changing the main areas of the agricultural and food industries. The study will discover how big data can help reduce waste in agriculture, food processing, and preservation. It will discuss the main hurdles and setbacks that prevent people from widely adopting these technologies, showing readers why something like this is not simple. Ultimately, by pointing out what is still unknown and what opportunities the future brings, this review will steer researchers, experts, and policymakers to focus on important and helpful new ideas for food systems.

Therefore, the following sections of this study are put together in detail below. Section 2 presents how the research was done, and the bibliometric information follows. Section 3 looks at how BDA aids in the key areas of agriculture, processing foods, and preserving them, and discusses the new technologies introduced and the obstacles to using them. The last section presents the research gap leading to the concluding remarks, the limitations, and the scope for future study.

2. Research methodology

This section provides a detailed description of all activities undertaken in our comprehensive review study. The PRISMA–2020 guidelines, which were laid out by Page et al. (2021) and Kahrass et al. (2023), were carefully followed during the research for this review. During the review planning phase, a pilot search was conducted to enable the authors to understand the literature, select keywords, choose a database, and establish inclusion and exclusion criteria. The main goal of a systematic review is to articulate the research topic accurately, enabling the selection of relevant papers. Fig. 1 shows the PRISMA flow chart of the study. This figure was created using a Shiny App (an online interface) as described by Haddaway et al. (2022).

2.1. Literature search protocol

The search terms, databases consulted, inclusion/exclusion standards used, publication types taken into account, and the assigned research period are all described in the literature search protocol. Based on this study, the pertinent keywords ("BDA, big data, agriculture, crop production, food processing, food preservation, and postharvest") relevant to the purpose of this investigation were utilised. A total of 6323 articles were retrieved from searches conducted in Scopus, Web of Science, and other reputable databases. The search was conducted on March 12, 2025, focusing on relevant publications published from 2015 to 2025.

2.2. Inclusion and exclusion criteria

We followed strict inclusion and exclusion criteria when conducting the review. Only full-text articles from peer-reviewed journals or review papers were included. Only articles written in English were included, while those in other languages were excluded. Additionally, papers unrelated to the discussion topic and those published in conference proceedings or editorial notes were excluded.

2.3. Data extraction

After applying the inclusion and exclusion criteria, which encompassed full access, 2609 articles were identified. The screening additionally identified duplicate articles, which were subsequently eliminated. Subsequently, the papers were evaluated for their titles and abstracts, resulting in the identification of 711 publications deemed pertinent to the study's focal areas: the impact of BDA in food processing and preservation. The articles were subsequently utilised for the bibliometric study. Following the bibliometric analysis, the study re-examined all 711 obtained documents for a comprehensive evaluation. Following a comprehensive examination of these articles, 181 were chosen for the review study. Notably, most of the reviewed articles were published between 2025 and 2015, reflecting the recent surge in research on BDA in food processing and preservation.

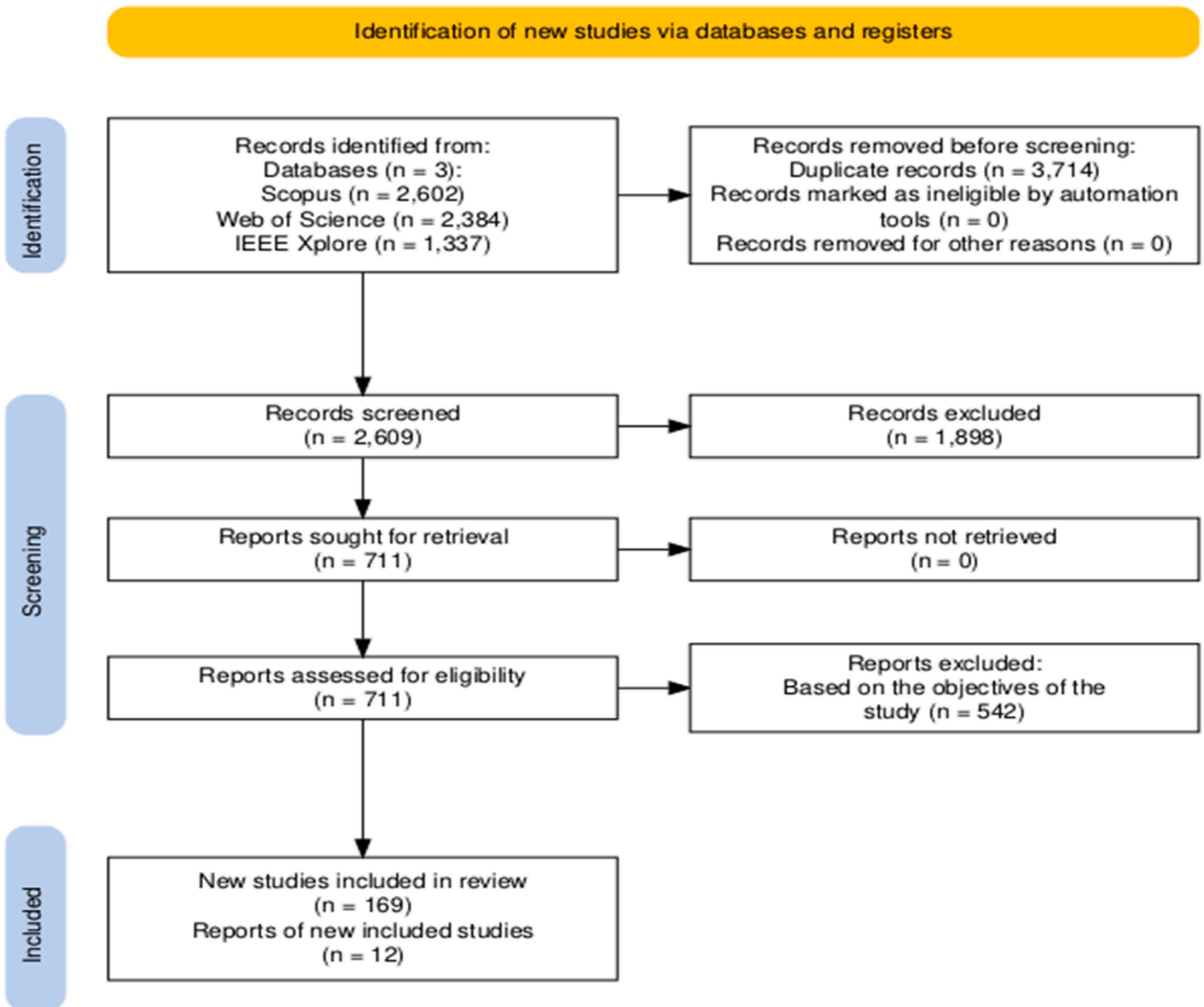

Fig. 1. PRISMA flow chart for the study.

2.4. Data analysis

Descriptive statistics were applied to analyse the literature. The VOSviewer and the Biblioshiny library in R-Studio were employed for bibliographic analysis. These two software applications are open-access web-based interfaces. The Biblioshiny library was used in R Studio for word analysis, trend analysis, and thematic map creation. VOSviewer was used to evaluate the keyword co-occurrence and do the cluster analysis.

2.5. Limitations

The study has limitations, including potential selection bias, overlooking non-English literature, and the authors' interpretation of data. The bibliometric analysis's limitations depend on the collected articles and data, and excluding related articles may affect the relationship and trends obtained. Additionally, the duration of the data collection process limited the study's scope. Future research could benefit from a more comprehensive analysis that includes a broader range of sources and languages to provide a more holistic understanding of the topic.

3. The bibliometric results

3.1. Main information and publication trend of BDA in agricultural practices, food processing, and preservation

Table 1 shows the bibliometric results of 711 scientific papers from 2015 to 2025, indicating a notable rise in academic interest in using BDA in agriculture, food processing, and preservation. The yearly growth rate is 3.63 %, indicating the increasing significance of data-driven solutions in tackling global issues such as food security, climate change, and sustainable agricultural practices. The research in this area is influential and widely recognised, contributing significantly to academic discourse and informing policy and industry practices. The knowledge base supporting this research area is vast, with nearly 50,000 references cited across the documents. Researchers are exploring a broad thematic scope, from machine learning for crop prediction to data-driven optimisation in processing and intelligent food storage systems. Collaboration is a hallmark of this field, with an average of over five co-authors per paper and 3269 contributing authors. The international collaboration rate of 37.27 % suggests that this is a global endeavour, with researchers from various countries working together to develop scalable and transferable solutions.

The balance between document types (546 research articles and 165 reviews) demonstrates that the field generates new knowledge and synthesises existing research to guide future developments. Original research drives innovation in technologies like AI-assisted food

Table 1
Main information of the examine articles.

Description	Results
MAIN INFORMATION ABOUT DATA	
Timespan	2015:2025
Sources (Journals)	260
Documents	711
Annual Growth Rate (%)	3.63
Document Average Age	4.41
Average citations per doc	66.83
References	49,991
DOCUMENT CONTENTS	
Keywords Plus (ID)	4735
Author's Keywords (DE)	2336
AUTHORS	
Authors	3269
Authors of single-authored docs	41
AUTHORS COLLABORATION	
Single-authored docs	41
Co-Authors per Doc	5.13
International co-authorships (%)	37.27
DOCUMENT TYPES	
Article	546
Review	165

processing and real-time preservation monitoring while review articles frame challenges and opportunities, offering frameworks for implementation and identifying research gaps.

3.2. The trends of publications over the year on the impact of BDA in agricultural practices, food processing, and preservation

The number of articles focusing on BDA in agriculture, food processing, and preservation has increased significantly from 2015 to 2025 (Fig. 2). From 2015 to 2018, the number of publications increased from 42 to 66, indicating the importance of BDA in the food and agriculture sectors. From 2019 to 2021, publications reached 78, indicating the rapid adoption of data-driven techniques in agri-food. The COVID-19 pandemic has accelerated the use of digital methods, leading to new developments in remote monitoring, automation, and supply chain strength (Kashem et al., 2024). However, Tripathy et al. (2024) reported a significant decline in the volume of academic publications in 2021, which was suggested to be due to the COVID-19 pandemic. From 2022 to 2024, the number of publications almost doubled, reaching 104 in the second year. This indicates that BDA technologies have made significant strides and are now well-applied in the real world. Researchers and practitioners focus on improving precision agriculture, creating AI-based food processing, and using smart preservation based on immediate and predicted information.

There were 60 papers published in 2025, more than some of the previous years. Additionally, this search was conducted in March, the first quarter of the year. The likelihood of further publications by the end of 2025 is higher. The rise in published research confirms the growing role of BDA in solving global problems such as food, climate, and supply chain challenges (Hasan et al., 2022; Koot et al., 2021). An increasing number of studies emphasise that BDA is not just about technology but also about improving agriculture and making it more sustainable.

Table 2
Most relevant sources and authors with their impact.

	h-index	g-index	m-index	Total Citation	Number of Articles
SOURCE					
Agricultural Water Management	21	36	1.909	1331	41
Plos One	17	34	1.545	1689	34
Computers and Electronics in Agriculture	16	27	1.455	2790	27
Proceedings of the National Academy of Sciences of the United States of America	15	16	1.364	3904	16
Agricultural Systems	13	13	1.3	503	13
Applied Sciences (Switzerland)	13	28	1.857	839	35
Soil and Tillage Research	10	12	1	623	12
Ecological Indicators	8	9	0.889	367	9
Field Crops Research	8	8	0.727	397	8
Nature	8	9	0.727	4630	9
AUTHOR					
Chen Y	5	9	0.455	83	10
Kumar S	5	7	0.5	1406	7
Liu S	5	5	0.714	193	5
Wang Z	5	5	1	134	5
Zhang X	5	6	0.455	2097	6
Zhao Y	5	6	0.556	105	6
Hoogenboom G	4	4	0.444	65	4
Li Y	4	4	0.5	279	4
Liu Y	4	5	0.5	166	5
Liu Z	4	5	0.4	106	5

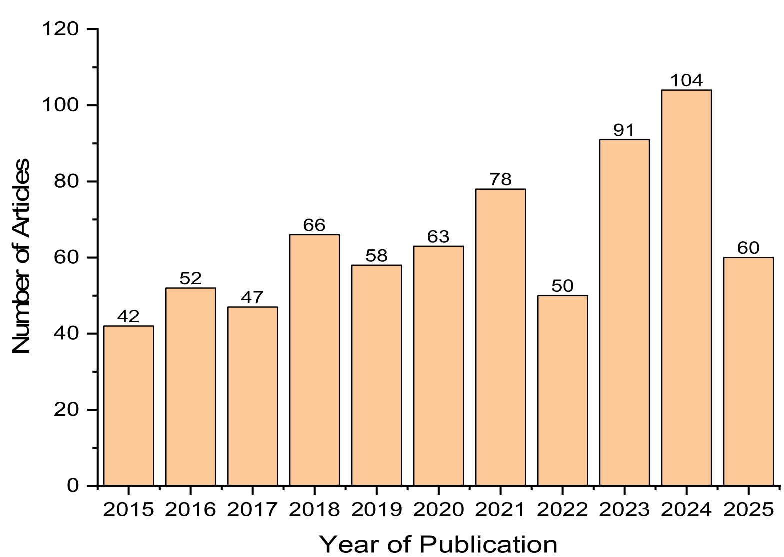

Fig. 2. Trends of publication over the evaluated year.

3.3. *Most relevant sources and authors with their impact*

Table 2 presents bibliometric indicators, such as h-index, g-index, m-index, total citations, and number of articles, for the top sources and authors contributing to BDA in agriculture, food processing, and preservation strategies. These metrics provide insights into productivity and impact, allowing for a deeper understanding of the research landscape and which journals and authors are leading its development (Tripathy et al., 2024). Agricultural Water Management is the most prolific source, featuring 41 published papers and an h-index of 21, signifying substantial output and significant citation impact. Plos One and Applied Sciences are also import and exhibit a harmonious blend of quantity and quality. Journals such as Computers and Electronics in Agriculture, Nature, and the Proceedings of the National Academy of Sciences of the United States of America are distinguished by their significant citation effect, with cumulative citation counts reaching as high as 4630 for Nature.

Other speciality publications, such as Soil and Tillage Research, Ecological Indicators, and Field Crops Research, provide significant contributions, albeit with comparatively lower citation metrics. According to their analogous h-index and g-index ratings, Zhang X, Kumar S, Chen Y, Liu S, Zhao Y, and Wang Z have demonstrated comparable contributions to the field. While Authors like Wang Z exhibit significant influence within a brief study timeframe (5 articles with five h-index and five g-index but one m-index), indicating that their recent contributions have been very impactful. Thus, this analysis has proven that BDA in agriculture and food systems relies on articles in prestigious journals and frequent contributions to specialized publications. With well-known and cited authors and articles, BDA is becoming more important in supporting sustainable and well-informed methods in the food industry. As the field continues to evolve, these sources and scholars will likely remain central to its development and application in real-world agricultural systems.

3.4. *Countries' scientific production and affiliations*

The distribution of scientific publications on BDA in agriculture, food processing, and preservation reveals global research activity and priorities. The top 5 countries, the USA (571 articles), China (551 articles), India (425 articles), Australia (150 articles), and Germany (131 articles), are the dominant contributors (Fig. 3a), focusing on precision agriculture, predictive modelling, and smart food processing technologies. However, this scientific production pattern highlights a global acknowledgement of BDA as a critical tool in transforming agriculture and food systems. These countries are leading research trends to have more robust funding, advanced technology infrastructure, and policies promoting digital agriculture and food innovation. This research spread across diverse regions underscores the global nature of the challenge, applying big data to address local, national, and regional agricultural needs and postharvest strategies. Fig. 3b shows the most relevant authors' affiliations. The China Agricultural University (35 articles), Sun Yat-sen University (29 articles), University of Florida (28 articles), Wageningen University and Research (27 articles), University of California (25 articles), and University of Bonn (24 articles) are leading institutions. Notwithstanding, many articles from all these institutions highlight the universal relevance and urgency of leveraging big data to transform agricultural and food systems worldwide.

3.5. *Characteristics of authors by keywords*

Fig. 4 shows the TreeMap for the available keyword combinations used in this study. Treemap visualisation presents an entire picture of the main topics and themes connected to research on BDA in agriculture, food processing and preservation. Crop production is the term that appears most often, showing that many studies focus on boosting harvests and farming efficiency using data. In addition, the text highlights agriculture and food supply, emphasising that developing technology is essential for keeping the entire food system running smoothly (Abiri et al., 2023; Liu et al., 2023). Themes such as "cultivation", "food security", and "climate change" show that more people are aware of the difficulties in agriculture and need to find efficient farm methods in changing weather and ensure enough food for everyone. Various terms related to sustainability, such as sustainable development and biodiversity, suggest that research focuses on productivity and keeping the environment safe.

It also points out the crucial importance of advanced technologies here. Forms of "big data", "machine learning", "artificial intelligence", and "data analytics" highlight the role of computer methods in guidance for agricultural practices (Rakesh et al., 2020; Teshome et al., 2024). "Decision making" is highlighted here, demonstrating how farming analytics assist farmers and stakeholders in making meaningful choices at the right time.. This area of research also concentrates on maize, rice and wheat, which are highlighted on the map, along with a special interest in "crop yield". This research uses big data to make essential food

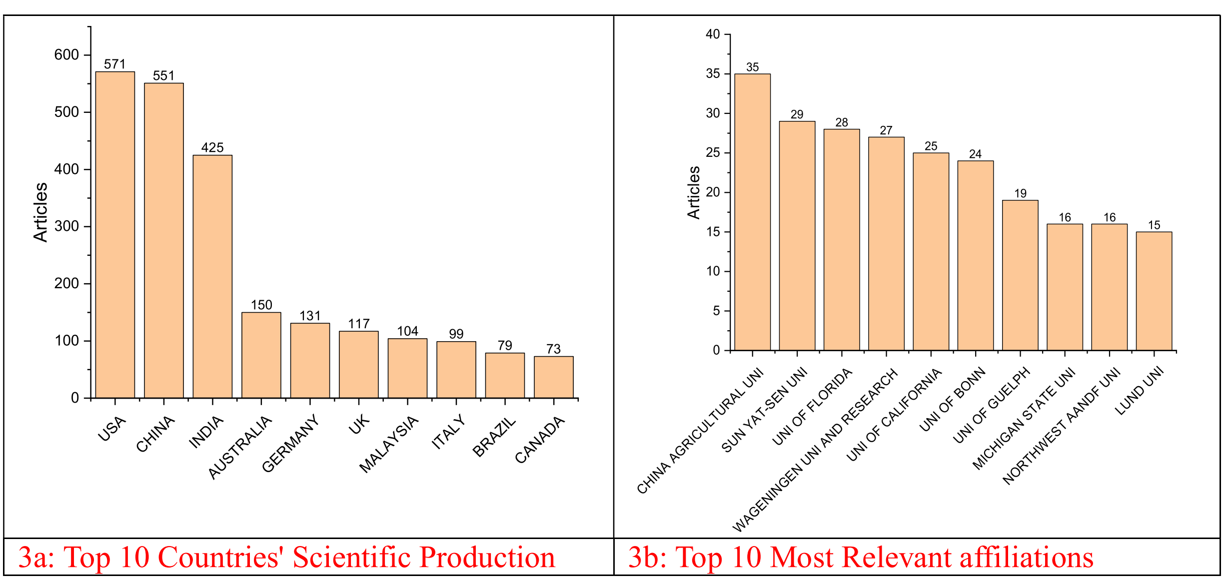

Fig. 3. Top 10 countries' scientific productions and authors' affilations.

Fig. 4. TreeMap for the available keyword combinations.

commodities more readily available (Zaborowicz & Frankowski, 2025).

Keywords about the food industry, production, and catering indicate that BDA research also covers food processing, distribution, and consumption, suggesting that it covers the whole food supply chain. For example, new fields such as nanotechnology and nonhumans focus on niche and innovative studies related to using unique materials or researching animal data in agriculture (Idamokoro & Hosu, 2022). Therefore, this graph demonstrates that BDA research in the agri-food sector ports is aimed at productivity, sustainability, using advanced tools and solving problems related to the environment. It shows that this field is moving forward by integrating innovations with help for farmers and the industry.

3.6. *Bibliographic coupling of journals, authors, and cooccurrence of keywords*

Bibliographic coupling is the connection between objects based on the number of references they share. It reveals similarities between the two works regarding documents, sources, authors, organisations, and countries (Tripathy et al., 2024). This study uses VOSviewer software to

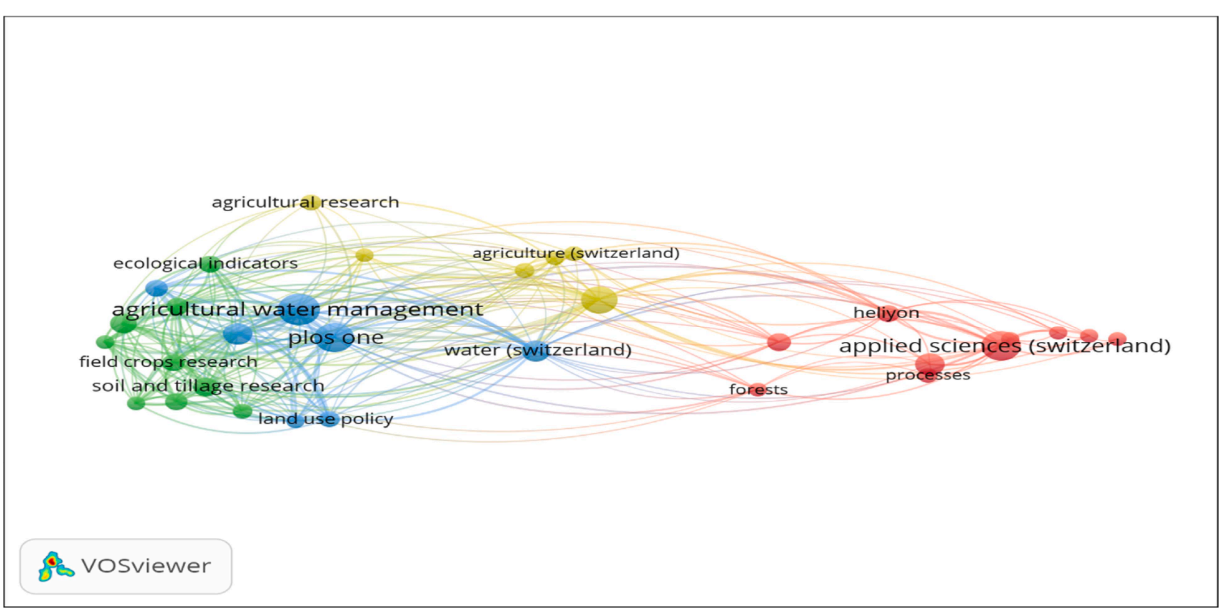

Fig. 5. Bibliographic coupling of journals.

analyse the bibliographic coupling of journals, authors, and countries, identifying instances where two publications are jointly referenced in a third article.

The references for journal papers on BDA in agriculture, food processing and preservation are displayed in Fig. 5. Out of 260 articles, only 32 meet the criterion, and the clusters of journal articles suggest one connecting through four nodes. Viewing all the data, the "Applied Sciences" journal, along with nine other publications like "Forests", "Heliyon", "Scientific Reports", "International Journal of Engineering and Technology" and others, is the one with the most links in the first cluster (identified in red). Green contains the second group and has Agricultural Systems as its leading journal, along with eight more, such as Agricultural, Ecosystems & Environment, and Agronomy for Sustainable Development. Furthermore, the last cluster, which is indicated in yellow, has "Agricultural Research" at its centre, together with five other journals. All the clusters play a key role in agriculture and food systems, and their networks demonstrate that improvements in one field can greatly influence changes in others. These four fields can use big data to enhance agriculture and keep food security constant.

Fig. 6 provides information about which countries are the most connected regarding BDA in agriculture, food processing and preservation. The minimum number of items for any nation's documents and citations was set to 5 to ensure that bibliographic coupling between countries meets the standard criterion. Only 56 out of 114 countries have met the necessary criteria. Also, the US, UK, Germany and the Netherlands were seen as the primary educational centres in these clusters. Still, their network ties indicate that where you are based in the world often shapes the research interests in BDA. Therefore, researchers from many countries should join forces to focus on improving energy efficiency.

Fig. 7 outlines the keywords used in BDA for agriculture, food processing and preservation and how they are related. A key part of this picture is a cluster (indicated in blue) that outlines the importance of BDA, machine learning, deep learning, AI, remote sensing, IoT and cloud computing in helping with data-based decisions in agriculture and food. A common set of keywords clustered in red centres on food security, sustainable development, global warming, crop production, and wildlife protection. It underlines how big data can help overcome world challenges related to food security and care for the environment, natural resources and biodiversity.

Another prominent thematic cluster, identified in green, focuses on managing resources and crops using data analysis to enhance crop outputs using just the right amounts of water and nutrients. The food processing and preservation constitute a unique cluster (depicted in purple) with the keywords food processing, food quality, food preservation, postharvest management, shelf life, sensors and supply chain management. Moreover, other clusters focus on applying big data findings to tangible actions and making these actions easy for users and policymakers so that farms and the entire supply chain can be managed efficiently and risks in farming can be mitigated.

The strong and visible relationship between the technological cluster (blue) and the crop and food clusters (green and purple) indicates that new technologies from AI and sensors are used in the farming and food industries. The clusters in decision-making (depicted in orange) and policy (shown in yellow) provide the link between innovation and, putting those ideas into practice and ensuring order. This co-occurrence cluster demonstrates the links connecting studies in BDA for farming, food processing and storage. It shows that its core technology relies on AI, machine learning, IoT and remote sensing. Moreover, technology is key in sustainable farming, crop and resource administration, and food quality monitoring. Along with these discoveries, this field combines bright ideas and science to help people grow more food, use resources efficiently and address sustainability.

4. Discussions

4.1. Background and significance of BDA in agriculture

Big data is generally characterized by three core dimensions: "volume", "velocity", and "variety". In the context of agriculture and food systems, "volume" refers to the large-scale data generated from sources such as sensors, satellite imagery, weather records, genomic data, and consumer behaviour data in the context of agriculture and food systems. 'Velocity' refers to the speed at which data is produced, analysed and worked. The 'Variety' points to the use of three types of data: structured (like databases), semi-structured (like XML) and unstructured (like images, videos, and social media. (Jin et al., 2024). This could mean combining soil data, drone footage, farmer records, and market trends

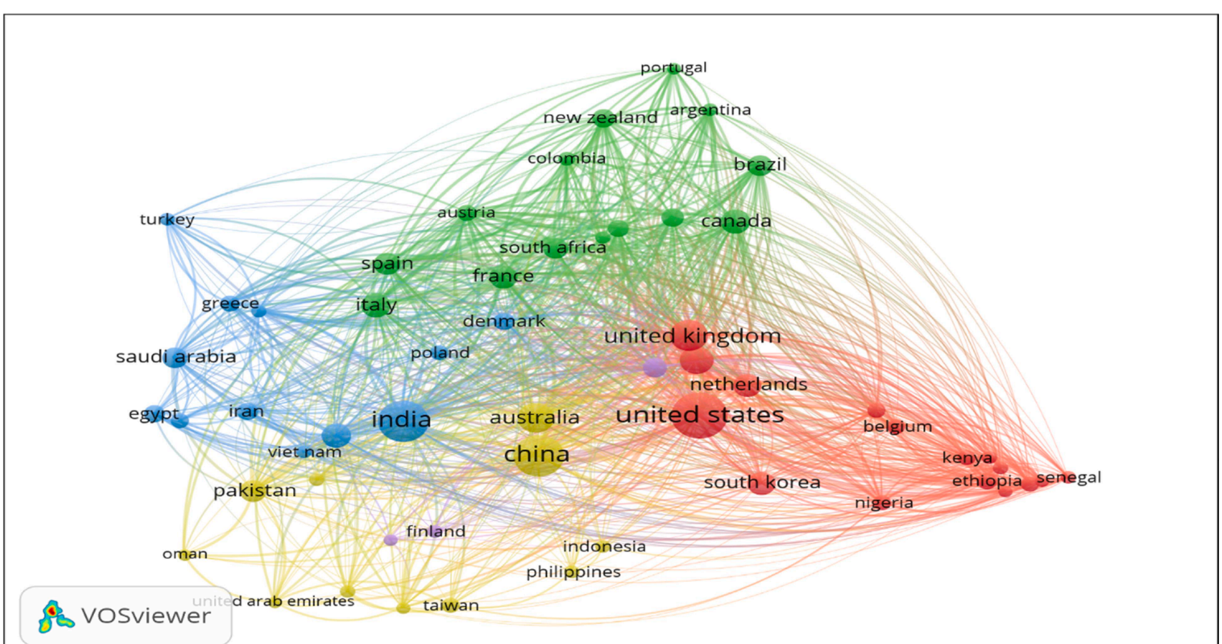

Fig. 6. Bibliographic coupling of countries.

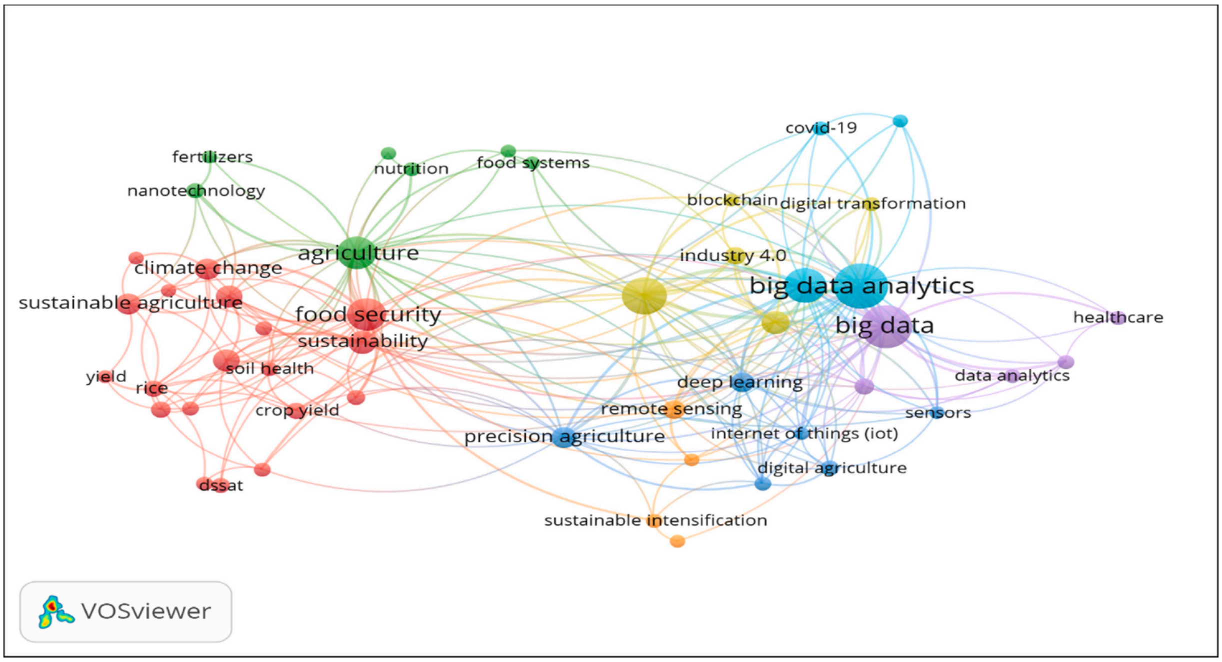

Fig. 7. Bibliographic coupling of keywords.

into one analytical framework in agriculture.

Contrary to common perception, big data extends beyond applying the three V's; it also entails extracting, processing, and interpreting valuable and actionable insights using a specialised array of tools and technologies. This now leads to incorporating further attributes (veracity and value) associated with big data. In certain instances, veracity is regarded as a fourth feature of big data, while in others, value is considered a fifth dimension, culminating in the 5 V's big data model (Demchenko et al., 2013). Fig. 8 delineates the defining attributes associated with the 5 Vs of Big Data. The 5 V's of Big Data are the essential foundations underpinning a concept that transcends mere processing and systematic collection of data sets that are excessively huge or complex for conventional data processing technologies to manage.

BDA encompasses various tools and innovative data processing techniques, especially with data collecting, storage, processing, and analysis (Theodorakopoulos et al., 2024). In contemporary agricultural systems, the tools and technologies encompass, but are not restricted to, data mining, machine learning, natural language processing, text mining, entity recognition, cognitive computing, cloud computing, data analysis, optimisation, and cyber-physical systems, among others (Ageed et al., 2021; Soni & Kumar 2022). BDA gives farmers and agribusinesses valuable insights based on data. This helps them make better decisions about choosing crops, when and how to plant, managing risks, improving operations, transportation, food distribution, and using resources like water and soil more efficiently. Big data continues to be an underutilised resource for enhancing the resilience, productivity, sustainability, and economic viability of global agri-food systems (Abiri et al., 2023).

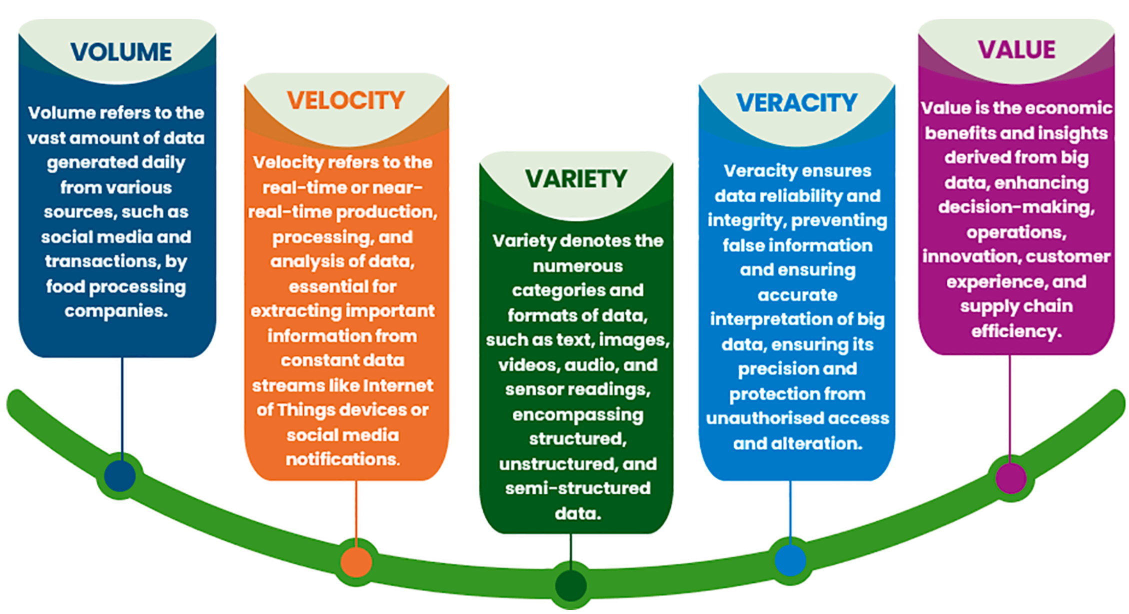

Fig. 8. The 5 V's Attributes of Big Data.

4.2. BDA in agricultural practices

BDA is revolutionising agriculture by optimising resources, enhancing crop management, and customising irrigation, fertilisation, and pest control using IoT sensors, satellite imaging, and weather forecasts (Fuentes-Peñailillo et al., 2024). Machine learning models provide precise crop output projections and early disease identification, outperforming conventional techniques (Akintuyi, 2024). Collaborative methods between farmers enhance group decision-making and crop results (Paul et al., 2022). However, issues like poor data quality, privacy concerns, and farmer training still exist. Despite these challenges, BDA offers practical tools for enhancing food security, efficiency, and sustainability in agriculture (Akintuyi, 2024).

Table 3 outlines the main facets of contemporary agriculture that big data is transforming and the data-driven strategies that help achieve better outcomes, increased production, and a sustainability-focused approach. In precision farming, crop monitoring, soil quality checking, and supply chain improvement, data is collected from different sources (e.g., sensors, weather stations, satellites) and used for analysis. The advantages are improved resource use, larger harvests, lower costs and more intelligent decisions. Tools such as IoT, various sensors and models for forecasting are listed for every application in the table. Using such technologies allows stakeholders in the agricultural field to choose actions that protect the environment, ensure enough food and sustain the industry.

4.2.1. Precision agriculture

Recently, interest has been resurgent in the applications of BDA within the agricultural sector, particularly in precision agriculture. The fundamental principle of precision agriculture is the application of innovative concepts and modern technologies to facilitate informed decision-making in optimising agricultural practices (Bhat & Huang, 2021; Osinga et al., 2022). For agriculture, several of the most powerful, transformative, and sustainable approaches directly or indirectly involve big data. BDA can be exploited in agriculture to provide actionable insights for enhanced decision-making, saving valuable resources, and reducing environmental impact.

Precision agriculture, otherwise known as site-specific farming, involves the application of information technologies on spatially variable fields to improve agricultural production practices, such as through tailored applications of water, nutrients, pesticides, and crop diagnostics. This typically uses concept elements such as variable rate technology, remote sensing, and Geographic Information Systems (GIS) to generate a "big picture" of the variability of the field, a global positioning system to identify the location accurately, yield monitoring for the very last process of precision agriculture, soil sampling to ensure the nutrients needed in each area of the field, as well as field data through the use of any sensors (Liu et al., 2021; Mani et al., 2020).

The fundamental aspects of precision agriculture consist of information, technology, and management, which are integrated to enrich production (Fig. 9). The information aspect involves gathering data on soil, weather, crops, and other variables affecting agricultural output. This data is collected from various sources, including sensors, drones, satellites, and land-based devices. After collection, the data is analyzed through sophisticated software and algorithms to produce insights. Farmers use these insights to make various decisions concerning their operations, such as when to plant, how much to fertilize, when to irrigate, and when to harvest. The technology aspect of precision agriculture embraces a considerable array of technological components that could be found inside a moderately-sized automotive shop, from the things one would find in a drone hobbyist's workshop to giant factory-like machines that can sow, tend, and harvest crops. The management aspect uses all this integrated information, mainly from the field, to efficiently operate a farm. Operate might be a poor word choice; these sophisticated (and almost always GPS-enabled) tools allow the development of a range of spatial datasets that display the differences in the field's performance or condition.

Precision agriculture's two most important aspects are location services and remote sensing technology, which ensure proper monitoring and reporting of crop status and problems. The three essential components of precision agriculture include deploying sensors around the farm, data analysis to assist farmers in decision-making, and automating specific tasks (Akhter & Sofi, 2022; Karunathilake et al., 2023). Various modalities of agricultural data collection include terrestrial data, vehicular data, and remotely sensed data. The earliest form of precision agriculture was yielding monitoring equipment that could record crop amounts at different places in the field. Since the adoption of precision agriculture, farmers have started to build up big data to improve future agricultural operations. With the push of technologies of the internet, mobile technology, wireless sensor network technology, GIS, and spatial semiconductor manufacturing, the development and extension of precision agriculture will be well promoted. Agribusiness and the food industry benefit from adopting a wide variety of tools offered by precision agriculture to optimise agricultural activities.

Big data intelligence helps understand when to plant and what variety to use, considering the climate change models, topography and soil types, and biotic and abiotic stress factors (Raza et al., 2023). It also helps in delivering good quality seeds and fertilisers on the farm over a space using big data and technological tools, optimising the uptake based on yield and the requirements of the plant around the 'spike' in

Table 3
Applications of big data in agricultural practices.

Area of Application	Description	Benefits	Examples / Tools	References
Precision Farming	Utilising data to enhance field-level management	Higher yield, lower input costs, and crop management tailored to a given site	Drones, Internet of Things sensors, satellite imaging, and GPS-guided equipment	Webber et al. (2019); San Emeterio de la Parte et al. (2023)
Crop Monitoring & Forecasting	Predictive analysis of crop, soil, and meteorological data	Improved crop health and disease management control	Remote sensing, NDVI analysis, AI-based forecasting models	Abiri et al. (2023); Chen et al. (2021)
Soil Health Analysis	Data collection on soil nutrients, pH, and moisture	Improved fertilisation techniques and environmentally friendly land usage	Soil sensors, lab analytics, digital soil maps	Hinge et al. (2021); Shahab et al. (2025)
Management of Irrigation	Monitoring soil moisture and water usage in real time.	Conserving water and enhancing crop quality.	IoT irrigation systems, weather data integration	Umutoni and Samadi (2024); Kamyab et al. (2023)
Optimisation of the Supply Chain	Monitoring and predicting developments in supply and demand.	Improved market access and less post-harvest losses.	Blockchain, RFID tags, logistics analytics	Hasan et al. (2022); Alsolbi et al. (2023); Lin (2024))
Management of Diseases and Pests	Early detection and modeling of outbreaks using environmental and crop data.	Minimised agricultural damage and decreased use of pesticides.	AI diagnostic tools, satellite-based disease tracking	Kaur et al. (2016); Achieng and Ogundaini (2024)
Market Intelligence	Examining market dynamics, consumer preferences, and pricing patterns	It helps in better decision-making for planting and marketing	Big data platforms, agri-market dashboards	Oncioiu et al. (2019); Yang (2024)
Climate-Smart Agriculture	Integration of climate data with farming practices	Adaptation to climate change, risk mitigation	Climate models, predictive analytics, GIS tools	Sebestyén et al. (2021); Ikegwu et al. (2024)

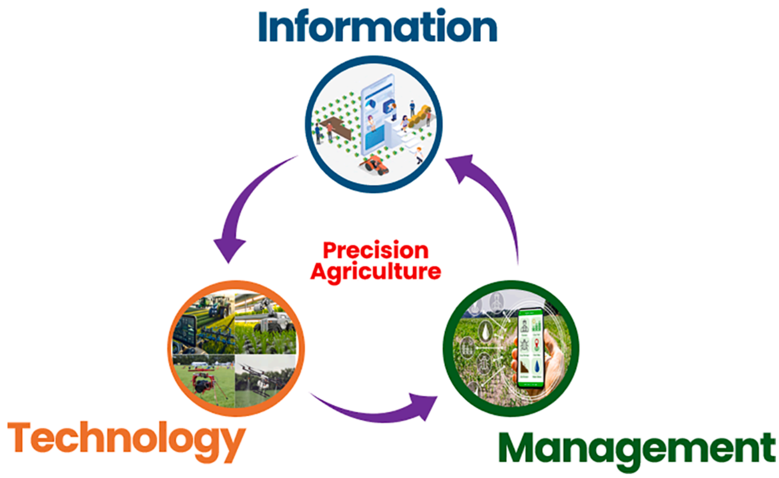

Fig. 9. The components of precision agriculture.

space, as well as the energy needs to invest in different sites, to deliver good quality yields.

4.2.2. Crop monitoring and management

BDA was developed relatively quickly, demonstrating its impact on scientific discovery and agricultural knowledge mining for efficient food production. Big data on agriculture is often used to describe an aggregate level of investment in information and communications technologies throughout the entire supply chain, for instance, by agribusinesses, farmers, food companies, water managers, and others (Akhter & Sofi, 2022; Osinga et al., 2022). Fig. 10 depicts the utilisation of big data in the monitoring and managing crops. Big data significantly transforms crop monitoring and management by facilitating real-time insights, predictive analysis, and optimised resource utilisation. It consolidates data from several sources to enhance agricultural decision-making, efficiency, and sustainability.

Big data is about acquiring or using large and complex information sets by digitalising a wide range of business activities and technologies. With the continuing digitalisation of agriculture and the rise of new biological, mechanical, and software solutions in digital farming, big data innovation in agriculture is now playing a key role in revolutionising crop planting to harvest cycles (Bhat & Huang, 2021; Kumar et al., 2022). One of the most significant innovators contributing to the intelligence of the farm is the emergence and advances in big data and, in particular, agricultural data analytics. Many stakeholders in the supply chain are now turning to big data. They are utilising sensors and other data collection and information delivery forms that hold a promise for the agricultural sector, which is the first of its kind.

Simply put, agriculture with BDA offers a future of farming and food (Bhat & Huang, 2021; Karunathilake et al., 2023; Osinga et al., 2022). Big data improves the accuracy, efficiency, and sustainability of crop monitoring and management, tackling global issues such as food security and environmental conservation. Its integration enables farmers to make data-informed decisions, assuring improved crop health and increased yield.

4.2.3. Livestock monitoring and management

Good animal health is essential to avoid any diseases that may occur at any time. Various methods, such as sensors or collars put on animals for tracking and record keeping, can significantly help detect any unusual behaviour of the animals. These technologies are also helpful for early detection and are used for monitoring body temperature to detect the signs of animal illnesses (Hamid & Ahmad, 2023; Hlimi et al., 2024). Tags are crucial on farms for assessing the weight, adherence to health

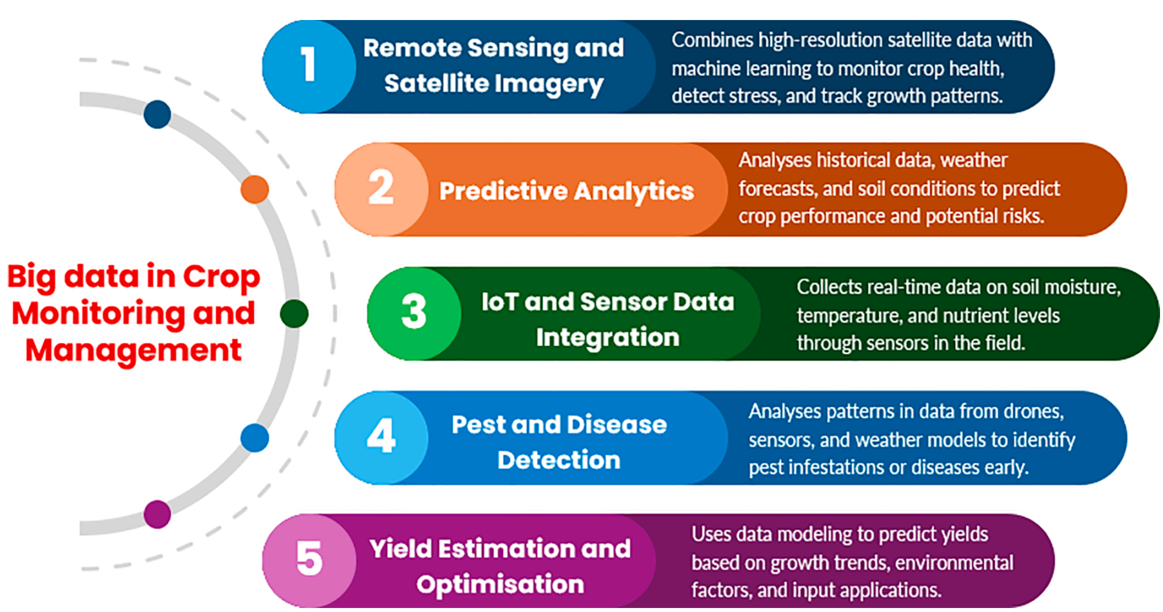

Fig. 10. Application of big data in the monitoring and management of crops.

protocols, temperature, and rumination of animals. Depending on the type, tags transfer data using different technologies. For example, Radio waves operate RFID tags, while NFC tags are meant for shorter-range communication. QR codes store data visually in a two-dimensional barcode, providing transparency from farm to consumer. Bluetooth tags, typically battery-powered, transmit data over 100 m, making them useful for real-time location tracking and monitoring environmental conditions (Kumari & Gaikwad, 2025). These tags are commonly used for cold chain management of perishable goods and tracking high-value assets. Fig. 11 illustrates the applications of big data in transforming livestock monitoring and management to improve production, health, and sustainability in animal husbandry. Farmers may make informed decisions that enhance animal welfare and operational efficiency by analysing extensive data gathered from many sources.

Numerous agribusinesses employ robotic systems for the automatic milking of cows. These systems are designed with integrated controls that permit ongoing monitoring of several vital tasks necessary for the operation of the farm, from the milking process itself to ensuring proper animal health and welfare (Lessire et al., 2020; Masía et al., 2020; Piwczyński et al., 2020). As robotic systems have become more widely adopted in the dairy farming industry, researchers have been swift to study their impacts on the many facets of the dairy operation to understand both the positive and the adverse effects (Edwards et al., 2020; Hansen & Stræte, 2020; Salfer et al., 2018). The survey conducted by Lage et al. (2024) examined the motivations and experiences of U.S. dairy farmers in adopting automatic milking systems, specifically in enterprises with seven or more milking units. The primary motivations for adoption encompassed labour expense reduction, improved bovine welfare, and enhanced herd performance. Wearable sensors are also available for dairy animals, affixed to their limbs, to gather data for monitoring animal health. Various wearable devices are available on the market that monitor advanced calving alerts, rumination activity, and temperature fluctuations indicative of oestrus. Documentation of multiple animal activities and individual production rates is also preserved, enhancing herd management (Alipio & Villena, 2023; Yu et al., 2024). Farm managers receive SMS notifications, which are crucial for disease detection by monitoring and attending to sick cows. Consequently, incorporating big data enhances livestock management's precision, sustainability, and profitability. It improves animal welfare while tackling issues like resource efficiency and disease outbreaks, meeting the global need for superior animal products.

5. BDA in food processing

The emergence of big data has provided food processing systems with a new dimension. Quality assurance in food is primarily conducted through tactile inspection, identifying physical defects in food, and utilising statistical methodologies called 'end-product testing.' This guarantees that the resulting meal meets the required standards. BDA mitigates food waste and enhances processing efficiency (Annosi et al., 2021; Bhat & Huang, 2021; Himeur et al., 2023). Data processing can diminish industrial waste, thereby leading to an increase in revenue. Efficient food processing will consequently enhance the competitiveness of the food processing industry. Conversely, optimising production procedures can reduce overhead costs, increase profitability, and ensure the company's financial stability and long-term growth. Organisations that can provide all critical elements of the food production chain are positioned to attract the most profitable consumers in the saturated global food market.

Addressing the numerous issues encountered by food processing organisations, inefficiencies in plant operations, and product quality is a fundamental catalyst for effective BDA. Food processing companies have challenges securing consistent supply, upholding quality, and achieving targeted production with reduced wastage rates relative to low-value items. Food processing companies encounter many complaints and product return requests stemming from inconsistencies in quality. Quality consistency poses a significant barrier since most multinational corporations acknowledge the lack of uniform quality across many batches of food items (Konur et al., 2023). Numerous food items require extended processing times; hence, data about their status must be monitored and documented for production control.

BDA can enhance the accuracy and consistency of data utilised for decision-making and forecasting in practical applications. Data scientists can develop a decision-making model to address various research, warehouse placement, and inventory challenges. Furthermore, several applications and software exist as instruments for BDA. In recent years, the principles of BDA have garnered significant attention due to their critical relevance and capacity for facilitating data-driven decision-

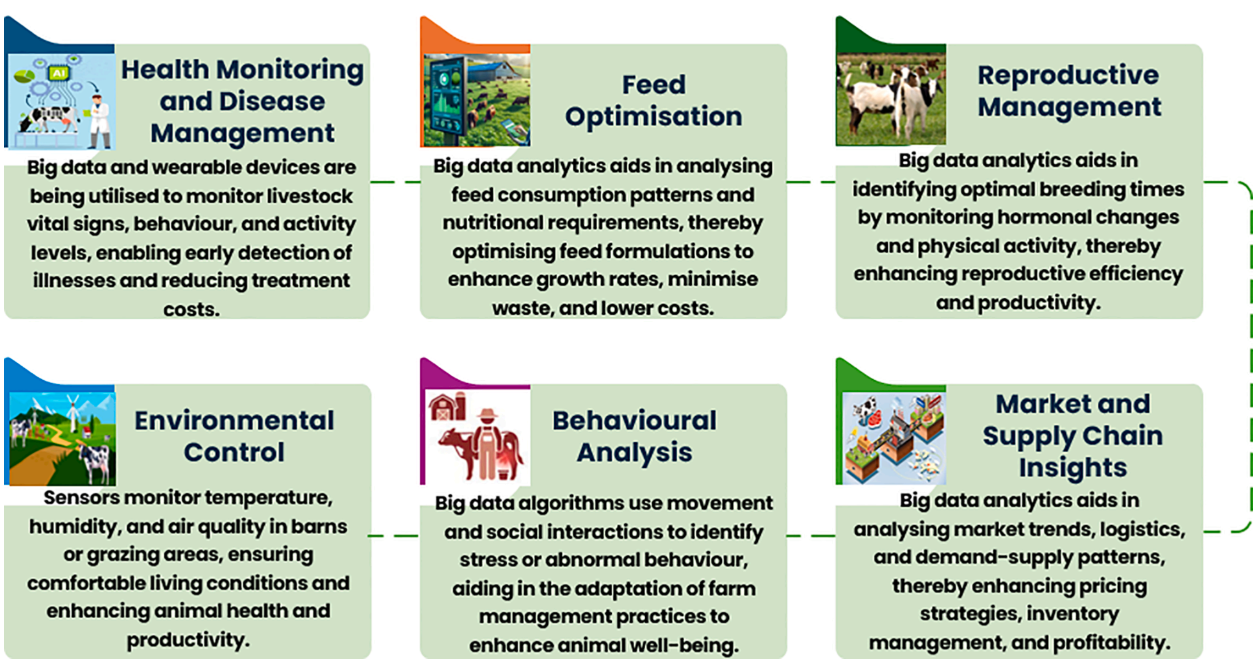

Fig. 11. Application of big data in the livestock monitoring and management.

making (Li et al., 2022). Real-time monitoring is important in food processing because it helps identify quality changes inside food materials during production processes. Quality by design demands real-time control tools to track and examine the quality changes in the food materials. High-technology applications such as artificial intelligence and near-infrared spectroscopy as real-time monitoring and controlling tools can transfer the traditional industry to innovative and intelligent industrial processes (Dodero et al., 2021; Hassoun et al., 2023a). Therefore, real-time monitoring and quality control of food processes to ensure food safety and sustainability are essential.

Fig. 12 depicts the various applications of BDA in revolutionising the food processing business through enhanced decision-making, process optimisation, and the elevation of product quality and safety. By analysing extensive datasets from many sources, enterprises may tackle issues such as waste minimisation, traceability, and customer contentment.

5.1. Areas of applications of BDA in food processing

5.1.1. Quality control and assurance

The significant public demand for safe and high-quality foods has rapidly become crucial to food processing. Monitoring the equipment, process, environment, and final product generates substantial data that can be examined for significant insights. Real-time monitoring is essential for identifying process deviations and facilitating immediate corrective actions (Liu et al., 2024). This service is beneficial in ensuring consistency when addressing the enquiries, "What is the correct process?" and "Is everything under control?" Comprehensive statistical models are constructed when there are low-frequency variance stakes or are challenging to validate empirically. A notable application of big data in quality control is real-time product surveillance. Data gathered from IoT sensors, cameras, and machinery during production monitors essential parameters, including weight, dimensions, temperature, and texture. This prompt input facilitates the identification of abnormalities on the production line, mitigating the danger of defective items and maintaining uniform quality.

An exemplary case is the application of hyperspectral imaging integrated with machine learning algorithms to identify food contamination and faults during processing (Nikzadfar et al., 2024). Hyperspectral imaging acquires an extensive spectrum of light beyond the visible range, offering comprehensive insights into food goods' chemical composition and physical characteristics. When integrated with BDA, this technology allows instant observation and assessment of quality in food processing lines. Also, in the nut processing sector, product safety and quality are essential, as they eliminate foreign objects like shells, stones, and other impurities from the finished product. The spectral signatures of the items allow hyperspectral imaging systems to detect the problems that conventional inspection techniques might miss. These problems might include minor faults or impurities (Ram et al., 2024).

An additional application that transforms the identification and forecasting of defects is the algorithms of machine learning that scrutinise historical and current production data to detect patterns that could lead to faults. The use of hyperspectral imaging in identifying aflatoxin contamination in peanuts has been demonstrated by Gao et al. (2021). By analysing spectral data, hyperspectral imaging systems can tell the difference between contaminated and uncontaminated peanuts and thus can eliminate dangerous items from the supply chain. This ability to predict allows businesses to set up proactive strategies to avoid downtime and cut waste.

Regulatory requirements demand that a certain level of quality and safety be assured across the food chain. The statistical processing software guides the company's testing plans by "enabling the input of different criteria levels and prioritising those of most importance." It also allows for analysis of data that comes from different sources. Four barriers resist the implementation of big data for quality assurance and control in agribusiness. These challenges involve issues with combining data, limited connections between sensory data, differences in sensory data, and the most formidable challenge of all, understanding real consumer needs (da Silveira et al., 2023; Osinga et al., 2022). To overcome these challenges, we need a complete plan that includes improving how we integrate data, enhancing connections between sensory data sources, reducing differences in sensory data, and using advanced methods to understand real customer needs accurately. Agribusinesses can utilise big data to enhance quality assurance and control methodologies by surmounting these obstacles.

5.1.2. Supply chain optimisation

The food industry is one of the biggest users of data analytics, and much of the analysis is concerned with improving the efficiency of the supply chain. Data are collected, ranging from climate condition information for farmers at the beginning of the chain to stock levels in retailers at the end and analysed to optimise resources at every stage of the

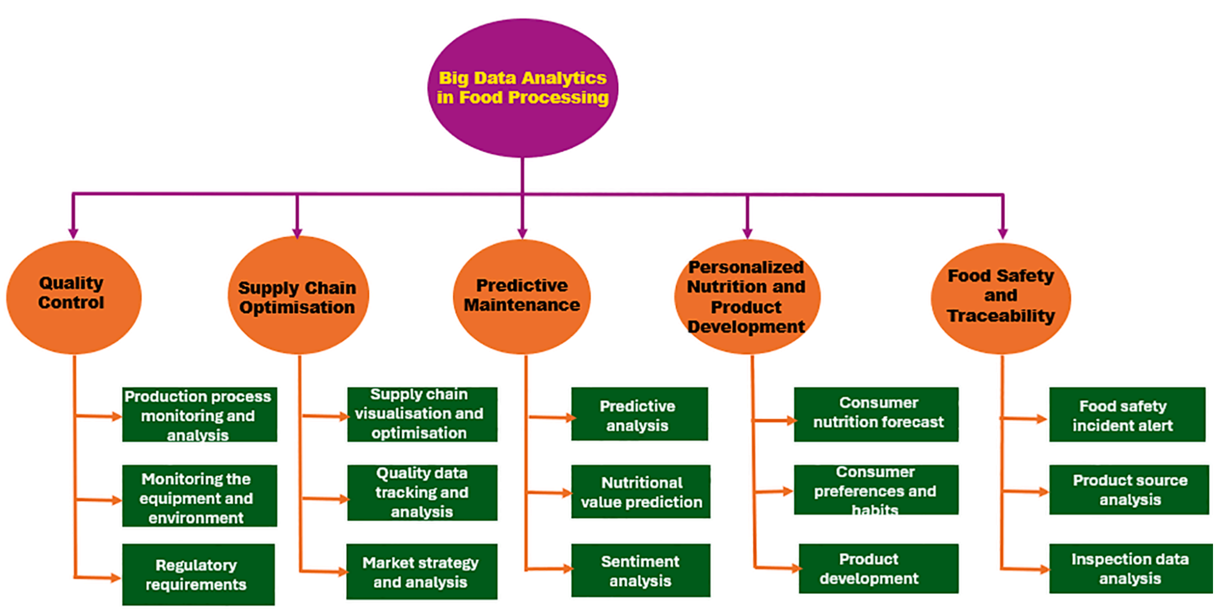

Fig. 12. Various applications of BDA in revolutionising the food processing sector.

chain and every activity within a stage (Bhat & Huang, 2021; Hassoun et al., 2023b). Like demand forecasting, inventory management is a significant benefit of big data. Analysing the data helps quantify the amount that needs to be held and where and when stocks should be placed to optimise high-quality and accurate products when consumers want them rather than unnecessarily holding stock (Bhat & Huang, 2021; Mashayekhy et al., 2022). Big data technology and analytics can be utilised to observe and improve understanding of traffic, weather, current conditions, route alterations, development, and distance. This information is included in a more advanced algorithm that calculates the time required to arrive at a delivery destination. Artificial intelligence and machine learning technologies can be utilised to enhance the prediction and provision of delivery timeframes.

An exemplary instance is the creation of the SLED (Shelf-Life Expiration Date) Tracking System, which employs machine learning algorithms to mitigate food waste and avert foodborne illnesses. The SLED system employs sensory observations to precisely evaluate food spoilage by juxtaposing expiration dates with distinct food attributes. It measures spoiling over time and trains machine learning models to forecast spoilage levels, offering consumers real-time insights regarding their food's freshness (Sonwani et al., 2022). A machine learning model determines spoilage by learning patterns and indicators associated with the deterioration of food products. This is done through data-driven training, where the model is fed many samples, some spoilt and some fresh, and learns to recognise features that correlate with spoilage.

Minimised waste is an additional advantage; for instance, big data may assess melons' optimal ripeness and dimensions to align with the tastes of specific major grocery stores, allowing producers to modify their fertilisation and cultivation techniques. Consequently, utilising big data enables the transparency of a supply chain, allowing for the direct tracing of quality and safety issues to their origins. This would enhance the management of safety issues, potentially diminishing the necessity for recalls.

5.1.3. Predictive maintenance

BDA may profoundly impact the food processing industry through many applications, such as predictive analytics, real-time data processing, and decision support systems for sustainable practices. Predictive maintenance is a targeted application of machine learning to evaluate machinery's lifespan or potential failure (Ayvaz & Alpay, 2021). AI-driven predictive maintenance solutions are becoming important in the industry, emphasising machinery health and reliability while aiming to reduce costs and minimise unnecessary effort (Abidi et al., 2022). AI-driven predictive maintenance oversees equipment and systems in real-time, facilitating the identification of anomalies and forecasting potential failures before their occurrence. This proactive approach reduces downtime, lowers maintenance costs, and ensures vessels remain seaworthy and efficient.

Kaur et al. (2023) and Michelena et al. (2024) noted that tailored data analytics for dairy processing can yield predictive, diagnostic, or prescriptive models that improve milk quality and processing efficiency. Dairy logistics must tackle milk variability; hence, analytics can improve understanding of this variability. Predictive analytics can alleviate the uncertainty of milk supply, and empirical evidence can validate this dairy logistics approach. Akhter and Sofi (2022) and Cravero et al. (2022b) noted that product predictive analytics and machine learning can significantly enhance decision support systems by tracking customer sentiments and suggesting items corresponding to consumer preferences.

Many businesses use the sentiment analysis technique to understand better how their customers' brand perceptions and behaviour are evolving. Data regarding product mentions on social media are extracted, aggregated, analysed, and visualised to produce statistically interpretable information that can inform business decisions. Sentiment analysis tools assist organisations in acquiring insights regarding real-time customer sentiment, customer experience, and brand reputation. Sentiment analysis is employed in the food sector to discern trends and identify popular products. The food business may utilise social media trends, such as hashtags, to pinpoint popular products for their forthcoming release. Consequently, with effective emotion research, organisations can enhance the design and development of future items with superior and acceptable attributes.

5.1.4. Personalised nutrition and product development

The field of nutrition is being transformed by BDA, which is providing unprecedented opportunities for individualised nutritional guidance and product innovation (Gami et al., 2024). As Lange et al. (2016) noted, big data provides possibilities for forecasting not just nutritional requirements but also health concerns likely to affect an individual based on a range of criteria. This makes it possible to set up not just preventative medicine in the traditional sense but also to install perceptive and preventative nutrition in an individual's life in a way that's tailored to them. Such an approach has the potential to avert traditional disease care problems pronounced in the modern medical model hyper-focused on illness and not wellness. Abeltino et al. (2022) noted that the analysis of an individual's food habits, physical activity, and body composition allows big data to develop personalised weight management programs that are more effective and durable. Thus, big data algorithms may provide highly customised nutritional plans by analysing an individual's genetic makeup, lifestyle habits, medical history, and gut microbiota. This can improve nutrient absorption, manage chronic illnesses, and boost overall health outcomes (Tsolakidis et al., 2024).

BDA facilitates the identification of consumer preferences and dietary requirements and developing trends, enabling food manufacturers to create products tailored to specific demographics and health issues (Hammad & Abu-Zaid, 2024). Big data can suggest personalised food goods to consumers according to their likes and needs, improving the shopping experience and fostering client loyalty (Guebsi et al., 2024; Javaid et al., 2023). Consequently, by analysing consumer feedback, sales indicators, and nutritional data, big data can improve product formulations to boost flavour, texture, nutritional value, and overall consumer happiness (Ding et al., 2023). Mariani and Wamba (2020) observed that BDA enables organisations to comprehend consumers' purchasing histories, preferences, and behaviours, thereby refining personalised product recommendations, pricing strategies, and promotions that enhance the shopping experience and foster consumer loyalty.

5.1.5. Food safety and traceability

BDA is transforming food safety and traceability by offering robust tools to analyse extensive data sets, identify possible hazards, and enhance the efficiency and efficacy of food safety management systems (Donaghy et al., 2021). Marvin et al. (2016) observed that by analysing historical data on epidemics, meteorological patterns, and various elements, big data algorithms can forecast possible food safety hazards and notify stakeholders proactively. This enables proactive strategies to avoid outbreaks and lessen their impact. Big data can reveal emerging food safety issues, such as new viruses or contaminants, by evaluating information from several sources, including scientific literature, social media, and consumer complaints (Donaghy et al., 2021).

An essential component of food safety, traceability entails the monitoring and tracking of food products across the whole supply chain. Traceability necessitates tracking and tracing food products along the entire supply chain, from manufacturing to consumption (Stanislawek et al., 2021). Traceability enables the rapid identification and recovery of food products that are compromised or dangerous. It alleviates the impact of foodborne illness outbreaks and protects public health (Stanislawek et al., 2021). The use of big data allows for watching food products in real-time across the supply chain, from production to consumption. This enables the rapid identification of contaminated products and provides for prompt recalls during outbreaks (Oriekhoe et al., 2024). Margaritis et al. (2022) stated that BDA integrate data from many sources, such as farm records, processing records, and transportation records, to view the food supply chain comprehensively. This capability makes them well-suited to recognise and help alleviate potential vulnerabilities.

For example, Kamath (2018) reported that IBM and Walmart joined efforts to implement a blockchain solution that helps make the supply chain of mangoes transparent and safer. The blockchain aids in the rapid tracking of the product's origin, transit conditions, and handling history, reducing the time needed to trace food sources from days to seconds. The system maintains an unchangeable record accessible to everyone from farm to store at every stage. This high transparency helps Walmart spot and remove contaminated batches quickly, resulting in less food loss and fewer health incidents.

The appearance of various analytical abilities has opened up new opportunities for assessing risks related to food safety and ensuring food safety for consumers. These opportunities are due to the growing number of tools that can analyse big datasets. Tools employed in big data are used to analyse the appearance of foodborne pathogens (the cause of foodborne disease outbreaks), which are detected in various locations (Deblais et al., 2020). These tools allow for a better understanding of food safety by using available knowledge alongside newly generated knowledge to analyse foodborne disease outbreaks as they are happening.

5.2. Application of BDA in food processing and preservation

BDA is a technology applied in food processing and preservation to improve food quality, safety, and sustainability. It has significantly enhanced industrial food preservation due to its ability to gather information from food preservation monitoring systems and analyse it using various methods. Control factors in the preservation process, such as storage duration, temperature and humidity levels, preservatives, techniques, and packaging types, provide guidelines for preservation methods designed for specific diseases (Bento de Carvalho et al., 2024; Konfo et al., 2023; Yang et al., 2022). Data analytics can enhance the strategic distribution of resources to manage the pandemic effectively and reduce its effects on food security (Balasundram et al., 2023; Tamasiga et al., 2023).

Table 4 underscores the revolutionary impact of big data and analytics on food processing, preservation and waste reduction. These texts illustrate a thorough framework in which BDA and developing technologies are pivotal in revolutionising food processing and preservation methods. The studies collectively highlight the capacity of these advances to elevate food quality, minimise waste, and raise the overall efficiency of food supply networks. BDA in food processing and preservation has made significant progress, but there are still gaps in its application within the food system. Standardised frameworks for integrating big data across stages are lacking, and research on the impact of BDA on long-term sustainability and ecological equilibrium is limited. Further study is needed to assess and use real-time data for dynamic food preservation and storage decision-making.

5.3. BDA in sustainable food systems

Sustainable food systems guarantee that all individuals have access to adequate, nutritious, palatable, diverse, and socially and culturally acceptable foods both presently and in the future (Fanzo et al., 2022). To accomplish this, they must be socially accountable, economically sustainable, and preserve or improve the environment. Sustainable food systems must adequately, nutritiously, and culturally satisfy the population; protect and preserve natural ecosystems and environmental integrity; provide consistently affordable and safe food of high quality for all individuals; guarantee equity and regional self-sufficiency both domestically and internationally; and generate livelihoods that enable individuals to reside within their communities (House et al., 2024).

Table 4
The revolutionary impact of BDA on food processing, preservation and waste reduction.

Application	Technique Used	Results/Impact	Reference
Optimising inventory control and pricing for perishables	Networked sensor data and dynamic supply chain decision-making	Enhanced food quality, improved consumer safety, and optimised revenue by correlating shelf life and pricing	Li and Wang (2017); Kayikci et al. (2022); Gligor et al. (2022)
Predicting customer behaviour and improving operational efficiencies	Utilization of sensor data and BDA in agriculture and biotechnology	Optimised plant responses to environmental changes and set the stage for data-driven food processing	Memon et al. (2017); GhorbanTanhaei et al. (2024)
Enhancing logistics and supply chain management	Integration of BDA and tracking technologies	Increased operational efficiency and better financial outcomes in food logistics and distribution	Govindan et al. (2018); GhorbanTanhaei et al. (2024)
Waste minimisation in the beef supply chain	Framework leveraging social media analytics, specifically Twitter data	Identification of consumer sentiments and operational inefficiencies, leading to reduced waste	Mishra and Singh (2018); Margaritis et al. (2022)
Real-time monitoring and predictive analytics	Application of IoT, machine learning, and Ubiquitous Computing Systems (UCS)	Highlighted the necessity for adaptive systems in manufacturing, enhancing monitoring and analytics	Aziz et al. (2019); Jamarani et al. (2024)
Transitioning to smart manufacturing in food processing	BDA within the Manufacturing Internet of Things (MIoT) and life cycle data management	Enhanced productivity and reduced waste in food processing industries	Dai et al. (2019); Belaud et al. (2019)
Food safety traceability and privacy protection	Blockchain technology integrated with privacy-focused frameworks	Strengthened traceability and ensured secure, transparent transactions in food supply chains	Tao et al. (2021); Lei et al. (2022); Xu et al. (2021)
Extracting actionable insights and addressing industry challenges	Analytical engineering for Big Data with focus on advanced technologies	Identified critical challenges such as data heterogeneity and privacy concerns, improving consumer trust	Rossi and Hirama (2022); Yang (2024)
Waste recognition and management in consumer-driven food systems	Machine learning algorithms for waste identification and production planning	Improved the resource efficiency and minimised environmental impact, ultimately creating a more sustainable and equitable food system	Onyeaka et al. (2023); Adewuyi et al. (2024)
Personalising dietary assessments and enhancing food quality	Integration of AI and Big Data with machine learning models	Improved food safety monitoring, personalised assessments, and real-time prevention of	Morgenstern et al. (2021); Min et al. (2023)

(continued on next page)

Table 4 (*continued*)

Application	Technique Used	Results/Impact	Reference
		foodborne illnesses	
Revolutionising food preservation and supply chain optimisation	Synergy of BDA and AI for food industry innovations	Enhanced food safety, optimised supply chains, and significant progress in addressing food preservation	Hasan et al. (2022); Ding et al. (2023)
Enhancing food security and streamlining supply chains	AI methodologies, such as support vector machines and convolutional neural networks	Improved food security, reduced waste, and streamlined supply chain management	Chen et al. (2024); Hassoun et al. (2025)
Dietary monitoring and food-related information retrieval	Integration of visual content, context, and external knowledge for food recognition and recipe analysis	Promoted better consumption practices, indirectly supporting food preservation efforts	Ding et al. (2023); Morales-Garzón et al. (2024)
Reducing food waste and improving quality	IoT technologies for real-time environmental monitoring and quality-controlled logistics	Enabled proactive measures to reduce spoilage and redirect products efficiently within the supply chain	Kolikipogu et al. (2025); Onyeaka et al. (2025)

The literature on BDA in sustainable food systems presents a complex landscape defined by the convergence of technology, agricultural methods, and socio-economic factors. Govindan et al. (2018) present an extensive study of big data applications in logistics and supply chain management in the context of smart farming. Their paper outlines the socio-economic issues related to integrating big data, emphasising its ability to enhance value creation within supply chains. Maya Gopal and Chintala (2020) discuss the limitations and prospects associated with big data in agriculture, emphasising the limited application of these technologies, especially among small-scale farmers. Their investigation uncovers many data sources and emphasises critical obstacles, including privacy issues and the risk of monopolies on important data. The authors promote collaborative initiatives among farmers to enable data exchange, therefore fostering the implementation of big data technologies that can improve production while maintaining sustainable practices (Hader et al., 2022).

Tao et al. (2023) emphasise the significance of the food industry, underscoring the critical role of big data in fulfilling customer demands for food quality amid rising malnutrition concerns. They assert that traditional food science has not met these needs, but big data offers innovative possibilities for scientific analysis and improved supply chain management. The bibliometric analysis demonstrates the increasing importance of big data in food safety and security, indicating that although the implementation of these technologies is still evolving, they possess the potential to revolutionise the food business. Lei et al. (2022) examine the amalgamation of big data and blockchain technology in food safety to improve traceability and mitigate food waste. Their research highlights the need for real-time data collection and analysis to ensure product quality and improve logistics. Employing IoT and AI technologies, they provide innovative solutions for informed decision-making in food production and distribution, addressing critical factors that contribute to food waste (Kravenkit & So-In, 2022; Lin et al., 2021; Treiblmaier & Garaus, 2023).

Min et al. (2023) expand upon these concepts by examining the potential of AI and big data to enhance agricultural productivity and sustainability. They highlight the capacity of sensory data from modern technologies to guide decision-making processes, improving crop yields and resource management. Their research illustrates how big data may be utilised to address significant challenges in agricultural productivity, such as soil health and climate resilience. Pan et al. (2023) provide Agriculture Data Management and Analytics (ADMA), a platform designed to tackle the complexities of data management in agriculture. Their findings highlight the necessity of adept data analysis to meet global challenges like climate change, stressing that data-driven approaches are essential for sustainable agricultural practices. Abiri et al. (2023) emphasise the importance of digital technologies in improving agricultural productivity and ensuring food safety. They advocate for the automation of farm operations and real-time monitoring to improve resource utilisation while addressing data privacy and security concerns. Ryan (2019) analyses the ethical implications of data use in agriculture. They emphasise the necessity of balancing data sharing with privacy protection, highlighting interdisciplinary collaboration's significance in improving sustainability in agricultural practices.

Ding et al. (2023) examine the amalgamation of AI and big data within the food sector, highlighting the advantages and obstacles associated with these technologies. Their analysis indicates that although the potential for enhanced productivity and safety is considerable, challenges such as data security and technical complexity persist as substantial obstacles to wider implementation. Gyamfi et al. (2024) propose a framework for Agricultural 4.0 that employs real-time data from diverse sensors to improve farm management and sustainability. Their findings suggest that technological integration can significantly affect food security and resource management, yet concerns over data privacy and accessibility persist. Bocean (2024) conducts a longitudinal study on the impact of digital technologies on sustainable food production in the European Union. His research underscores the imperative of integrating social, economic, and environmental considerations into agricultural practices, advocating for education and technological accessibility to facilitate a transition to more sustainable food systems.

These studies collectively illustrate the revolutionary capacity of BDA in sustainable food systems while emphasising the problems that must be addressed to harness this potential effectively. The identified problems and research challenges in using BDA in food supply chains show that most studies focus on the early parts of the supply chain. These include aspects like inventory levels, management of tasks, and meeting schedules, as well as key players like suppliers, customers, producers, and warehouses. However, the amount of research focused on the last phases of the supply chain is limited; resources such as logistics and customer process management, along with relevant stakeholders, are increasingly deficient. The report examines the ramifications of utilising BDA in food value chains and proposes future research avenues for each pertinent stakeholder. Specifically, concerns about governance, control of contextual impacts, and their consequences for trust throughout the food chain are highlighted. Consequently, using big data can be advantageous for smaller enterprises if they exploit market niches and cultivate client confidence in the quality and safety of their products. A practical approach to accomplish this is by employing transparency strategies.

5.3.1. *Benefits of BDA in sustainable food systems*

BDA significantly enhances sustainability in food systems, providing several advantages that tackle environmental, economic, and social issues. Fig. 13 emphasises the benefits of BDA in sustainable food systems. According to Musanase et al. (2023), big data-enabled precision agriculture enables farmers to optimise their practises using real-time data on soil conditions, weather patterns, and crop health. This results in enhanced efficiency in utilising water, fertilisers, and pesticides, diminishing environmental impacts and augmenting yields (Çakmakçı et al., 2023). Moreover, BDA can detect and eradicate inefficiencies in the food supply chain, reducing waste at all stages, from production to consumption.

Cravero et al. (2022a) noted that analysing climate data via big data might enhance the forecasting of climate change effects on agriculture

Fig. 13. Advantages of BDA in sustainable food systems.

and guide the development of climate-resilient agricultural practices. This is crucial for ensuring food security in a changing climate. Furthermore, big data can inform consumers about their food selections' environmental and social ramifications, promoting more sustainable consumption behaviours (Chandra & Verma, 2023). While it is crucial to tackle issues like data quality, privacy, and accessibility, the advantages of BDA in fostering a more sustainable food system are significant.

BDA offers a substantial opportunity to improve sustainable agricultural systems (Rejeb et al., 2022; Singh et al., 2024). By leveraging data, we can enhance resource efficiency in agriculture, reduce supply chain waste, and alleviate food production's environmental consequences. In conclusion, implementing a sustainable food system strategy can promote the rejuvenation of the planet and humanity. This philosophy emphasises compassion for human health and offers numerous opportunities for the well-being of affluent persons and the environment.

5.3.2. Challenges and opportunities in implementing BDA in the food system

BDA is extensively utilised across the food chain, from crop production and harvesting to processing and shelf-life extension. However, implementing new technology often poses a complex task, and acceptance rates usually demonstrate significant variety. Deepa et al. (2022) and Himeur et al. (2023) found that problems like high costs, worries about data privacy and security, and the need for sufficient physical infrastructure limit the use of big data in food and agriculture. Therefore, stakeholders must confront these challenges by concurrently developing the tools and incentives required for thorough implementation across the food and agriculture sectors. Challenges persist in implementing big data technology into agriculture and food systems. Data security and privacy concerns may inhibit stakeholders from disseminating useful information (Lee & Ahmed, 2021). Moreover, there is a necessity for physical infrastructure, like broadband connectivity, between areas to facilitate data-sharing activities. Resistance to change within conventional agricultural systems may impede the adoption of big data technologies, particularly at the farm level (Keshta & Odeh, 2021; Lee & Ahmed, 2021).

Although BDA offers excellent potential for increasing innovation and improving the efficiency of food systems, there are unique challenges in terms of quality, organisation, and ease of access that must be rectified before we can fully utilise the vast amount of data generated in these systems (Ding et al., 2023; Ikegwu et al., 2022). A limiting factor is access to data. Osinga et al. (2022) reported that not all producers and other stakeholders can collect or connect to the current BDA infrastructure. Further, a significant barrier often exists to integrating, analysing, and finally interpreting the vast amount of food data. The underlying infrastructure, such as equipment and software, often cannot communicate or be integrated.

Lastly, consideration must be given to the human side. Many individuals lack an understanding of BDA, how it can be implemented, methods for analysing this and other big data infrastructure systems, and whether they have the necessary workforce for effective big data utilisation (Lutfi et al., 2022; Mikalef et al., 2021). A shift is required in workforce preparation to educate future employees on the properties of BDA, including database knowledge, statistical and programming skills, and advanced IT skills. Several obstacles exist in the industry for integrating BDA into food systems, which will need to be addressed to see real adoption.

Nevertheless, the application of BDA also presents numerous opportunities for utilising farmer-focused technology, methodology, and software strategies to harness the additional production, environmental, and social benefits through this technology and create a precise approach (Bhatia & Kaur, 2024; Prodhan et al., 2024). Deepa et al. (2022) and Naeem et al. (2022) noted that access to and maintenance of data quality are primary concerns for BDA, but also for required information data aggregation, sharing, and standardisation systems such as collaborative research initiatives, databases, and knowledge systems. Moreover, cultivating public-private partnerships may enhance progress towards these desirable agricultural objectives. These collaborations can enhance the exchange of resources, talent, and technology to increase data quality and accessibility for BDA in agriculture. Cooperative initiatives among governmental bodies, research organisations, and commercial enterprises can produce more efficient and sustainable agricultural practices.

Opportunities for BDA also encompass enhanced decision-making and superior forecasting of market trends and behaviours. Consequently, by utilising data, players in food systems are innovating to improve compatibility and increase acceptance (Keshta & Odeh, 2021). Chaudhuri et al. (2023) observed that governmental and industry-backed consortiums are developing platforms for collecting and analysing massive data sets, aiming to disseminate outcomes to provide value to stakeholders. By tackling significant obstacles through policy and infrastructure development, it is feasible to support these projects and promote collaboration among stakeholders across various segments of the food chain.

5.3.3. Opportunities for innovation and growth

The increasingly affordable data analytics technologies are creating sizable opportunities to enhance agricultural production, food processing, and preservation efficiency and productivity. Advanced data analytics can potentially improve operational efficiency, optimise inputs, harden equipment, and forecast outcomes (Boppiniti, 2022; Shahin et al., 2024). Moreover, data analytics can create innovative business models, offer insights into supply chain efficiency, and reveal new monetisation opportunities (Ofulue & Benyoucef, 2024). Data analytics insights can be utilised to enhance operational efficiency in corporate systems, work settings, and agroecological functions (Rejeb et al., 2022). Therefore, for disruptive technologies to take advantage of the sector, it is crucial to promote extensive coalition-building among all pertinent stakeholders in the global food system. Shawki and Schnyder (2021) emphasised the industry's need to prioritise future knowledge and new frameworks over conventional ethical and financial principles.

Consequently, any enterprising entrepreneur or leader will assert that modernisation and research and development are essential to a firm's success. The integration of artificial intelligence and BDA to acquire agronomic knowledge or cultivate local produce will establish the norm in 21st-century agriculture and food systems; AI has the potential to revolutionise global farming and harmonise productive and renewable domestic and international practices (DayioĞLu & Turker, 2021; Sharma et al., 2023). With technological advancements, society will see substantial growth. Artificial intelligence and BDA have the potential to revolutionise agricultural practices, leading to increased sustainability and higher efficiency. By embracing these technological advancements, the agricultural industry can adapt to confront the problems of the 21st century and beyond.

6. Ethical and legal considerations

As we advance the capabilities of BDA in agriculture and food systems, some ethical and legal considerations must be addressed. Special attention should be paid to data privacy, security, and algorithmic fairness. In recent years, there has been a significant increase in the volume of sensitive data collected in agriculture and food systems. Companies should, therefore, seek to comply with the rules, standards, and ethical principles specifically developed for BDA, which refer, in general, to the trusted use and fair processing of data in alignment with universal human rights and underlying principles. Consent in data collection and transparency about such collection are also particularly relevant in ethical considerations for precision agriculture and data mining in agriculture and food systems.

6.1. Data privacy and security

The massive amount of data gathered during BDA includes personal data and sensitive information. Information breaches can reveal information to unauthorised groups that can affect the data owner and others. For instance, disclosing personal information such as age, economic status, and health can lead to discrimination; therefore, big data comes with serious concerns about data privacy and security. Privacy and data protection legislation restrict the use of personal and other sensitive data attributes in data collection, analytics, visualisations, and data sharing (Hasal et al., 2021; Lutfi et al., 2022; Ogbuke et al., 2020). Legislation and regulations outline measures for organisations in the food systems for adequate data protection and processing to ensure compliance by processing personal data with data protection and digital security features incorporated into the design and development of various sensors and digital solutions employed in precision agriculture and food systems.

In the context of digital solutions currently used in most food systems, sensitive data, including personal consumer and commercial data, are being collected. Data privacy, handling, and security requirements are heightened when dealing with personal data. Data stewards have an ethical responsibility to ensure the data's safety, security, and privacy (Lee & Ahmed, 2021; Quach et al., 2022; Yap et al., 2021). Responsible stakeholders should reasonably safeguard digital solutions against well-known security threats using methods and technologies such as encryption methods, secure storage, data anonymisation, and improved practices by ensuring that sufficient technical solutions are employed to guarantee high cybersecurity. Inau et al. (2023) and Aksenova et al. (2024) indicated that a summary of data management encompasses monitoring the data from its creation to its utilisation in research and dissemination. This will also encompass efforts to safeguard sensitive data. Authorised access to data must be defined with explicit permitted usage. To protect commercial and consumer data, the entry of sensors, smartphones, and other recording-capable electronic devices may be restricted in specific regions.

6.2. Regulatory compliance

Regulatory compliance comes with interdisciplinary significance in this era of data-driven optimisation and automation in food systems. Authenticity in food prediction ensures business value data integrity and sovereignty. Respecting the data and making it anonymous enough while making it available securely to the computing industry is also a challenging but unavoidable strategy in the present era. Many regulations demand the safeguard and sovereignty of the operational data for a respective time, after which the data has to become obsolete to some extent, ensuring historical association and traceability. In some scenarios, the data has external intellectual property impacts where the data and its products or predictions have third-party rights, and this intellectual property protection is possible only through compliance and certifications.

Several regulations encompass data security and sovereignty. The responsibility of using and protecting data and registration includes many layers of legislation and compliance. The major data protection regulations required for the food digital supply chain and security in general are the European Union's GDPR (Khazaei & Arabsorkhi, 2024; Nurgazina et al., 2021), the United States HIPAA (Oriekhoe et al., 2024; Wilgenbusch et al., 2022), and Japan's Act on Protection of Personal Information in the Private Sector and Act on the Use of Numbers to Identify a Specific Individual in Administrative Procedures (Habuka, 2025; Reis et al., 2024; Wang et al., 2022). These regulations seek to guarantee the confidentiality and security of personal data while providing directives for managing and processing sensitive information. Adherence to these regulations is essential for organisations in the food digital supply chain to prevent legal repercussions and sustain customer confidence.

Consequently, consumers are demanding how and where technology and data aspects are linked and used and where the data protection regulations are demanded and imposed in the IoT environments. Central regulatory bodies regulate food-related data, workflow, governance, and compliance. Information governance is the primary area responsible for data privacy and protection. Symbols and policies are rendered, and different subsectors and foods formulate and adhere to standards. However, the methods and strategies are liable to be continuous.

7. Future trends and research directions

Adopting BDA in agriculture and food systems can be a game changer as we move towards a planet with over ten billion people. Data-driven techniques are expected to offer a long list of benefits in agriculture, such as monitoring soil nutrients and understanding soil types, and the ability to predict food production, food safety, climate change impacts, crop diseases and pests, and optimal product storage and shipment durations. Moreover, such techniques offer a clear pathway to faster food quality inspection and shorter supply chains. Precision agriculture is increasingly being adopted by companies seeking to increase crop yields while reducing the environmental footprint and, hence, developing more sustainable practices. When it comes to food processing, adopting analytics is expected to optimise the operations of manufacturing and supply chains. Adopting these techniques, especially IoT-related technologies, can help predict food production failures, optimise plans for existing facilities, contribute to new product or facility designs, manage supplies, and help reduce waste.

IoT devices and technologies like Low-Power Wide-Area Network (LPWAN), Home Node B (HNB), Integrated Pest Management (IPM), advances in communication technologies, software platforms, and mobile applications can be potential research directions. In addition, implementing BDA, data-driven models, and artificial intelligence applications can pave future research directions in agri-food extension sections. There is a need for research in the interdisciplinary collaboration of computer science, informatics, food science, agriculture, and environmental studies to integrate big data in sustainable food systems better.

Research is required to integrate big data with resilience thinking, resilience measurement, and identification of the trade-offs between the food system elements and resilience. Applying big data to consumer behaviour to encourage healthier, more environmentally friendly diets, especially those that could reduce greenhouse gas emissions. Sharma et al. (2022), Xu et al. (2022), Akhter and Sofi (2022), and Cravero et al. (2022b) noted that big data can be used to develop reliable models to predict the impacts and probability of undesirable events, including trade relations, lack of adaptive decision-making by large and small companies, investments in energy infrastructures, and other technologies for mitigation. Research is needed to study the maximum or minimum benefits for society relative to big data availability and future flexibility, thus raising the questions of data monetisation, privacy, and agency control. Future research should focus on developing robust governance models that ensure transparency, data privacy, and equitable access to technological resources across the food supply chain.

It is accepted that big food data could be obtained from internal information sources and shared with customer-end and other agencies for real-time food quality and safety management and rapid adoption of preventive measures in multi-step food processes. There is a view to promote a more efficient supply chain and positively impact total inventory management. BDA today is believed to be available to suppliers, retailers, and final consumers who use their mobile devices to form a sentiment analysis using BDA of all voice social media networks for reference after consumption or use. This will enable the supply chain to respond to demand and keep as little safety stock inventory as possible. This calls for more research in the multidisciplinary field of food-data science, where universities meet with industry researchers for food to meet industry needs and further broaden this idea of big data for the food supply chain. Since many of the big data opportunities in sustainable food systems are at the intersection of big data applications and food systems, more research is needed and fast.

8. Conclusion

There is a view to promote a more efficient supply chain and positively impact total inventory management. BDA today is believed to be available to suppliers, retailers, and final consumers who use their mobile devices to form a sentiment analysis using BDA of all voice social media networks for reference after consumption or use. This will enable the supply chain to respond to demand and keep as little safety stock inventory as possible. This calls for more research in the multidisciplinary field of food-data science, where universities meet with industry researchers for food to meet industry needs and further broaden this idea of big data for the food supply chain. Since many of the big data opportunities in sustainable food systems are at the intersection of big data applications and food systems, more research is needed and fast.

The review also points out the need for systems that combine all areas, grow in scale and make real-time information sharing, analysis and automation available for the whole food industry. It is crucial for data privacy and how it is managed to be considered for trust and fairness. Given this, researchers and organisations from multiple fields should join forces to fill existing gaps and speed up the use of big data in agriculture. Therefore, the successful use of BDArequires a balance between technological improvements and the concerns of society, the economy and the environment. Focused work with big data could lead to an agri-food system that supplies the world with the safe, healthy and accessible food it needs. The recommendations for further studies based on the review findings are:

1. Implement interoperable platforms for seamless data collection, real-time analysis, and agricultural and food supply chain automation to enhance stakeholder efficiency and decision-making.
2. Invest in digital infrastructure, especially in rural and underdeveloped regions, to ensure equitable access to big data technologies, particularly among smallholder farmers, for reliable connectivity and data processing.
3. Encourage studies on big data analytics impact on postharvest processes, food preservation, and consumer behaviour to fill knowledge gaps and enhance end-to-end food system efficiency.
4. The initiative aims to foster collaboration among government agencies, research institutions, technology providers, and agribusinesses to enhance innovation, standardise best practices, and support scalable solutions.

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Availability of data and materials

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Consent for publication

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Ethical statement - studies in humans and animals

This article does not contain any studies involving human participants performed by any of the authors.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article.

CRediT authorship contribution statement

Jelili Babatunde Hussein: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Tilahun Seyoum Workneh:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Alaika Kassim:** Writing – review & editing, Validation, Investigation. **Khuthadzo Ntsowe:** Writing – review & editing, Visualization, Investigation. **Sileshi F. Melesse:** Writing – review & editing, Validation, Supervision, Resources. **Hany S. El-Mesery:** Writing – review & editing, Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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